Project: Vulnerability Assessment and Penetration Testing

Assessed Systems:

- Windows Machine – 172.31.32.114
- Debian Linux Machine – 172.31.42.236

Tools Used:

- Zenmap (Nmap GUI)
- Nessus Vulnerability Scanner

Step 1: Performing Information gathering and Recon

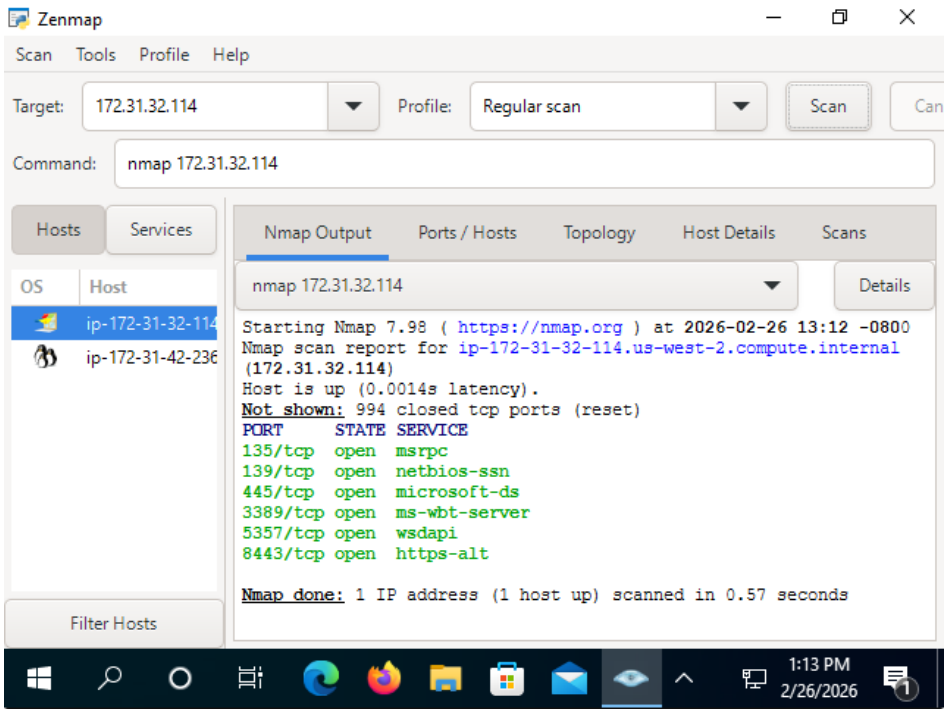
1.1 Windows Regular Scan

Command Used:
nmap 172.31.32.114

Open Ports Identified:

- 135/tcp – MSRPC
- 139/tcp – NetBIOS
- 445/tcp – SMB
- 3389/tcp – RDP

Observation:
The host is active and multiple services are exposed to the internal network.



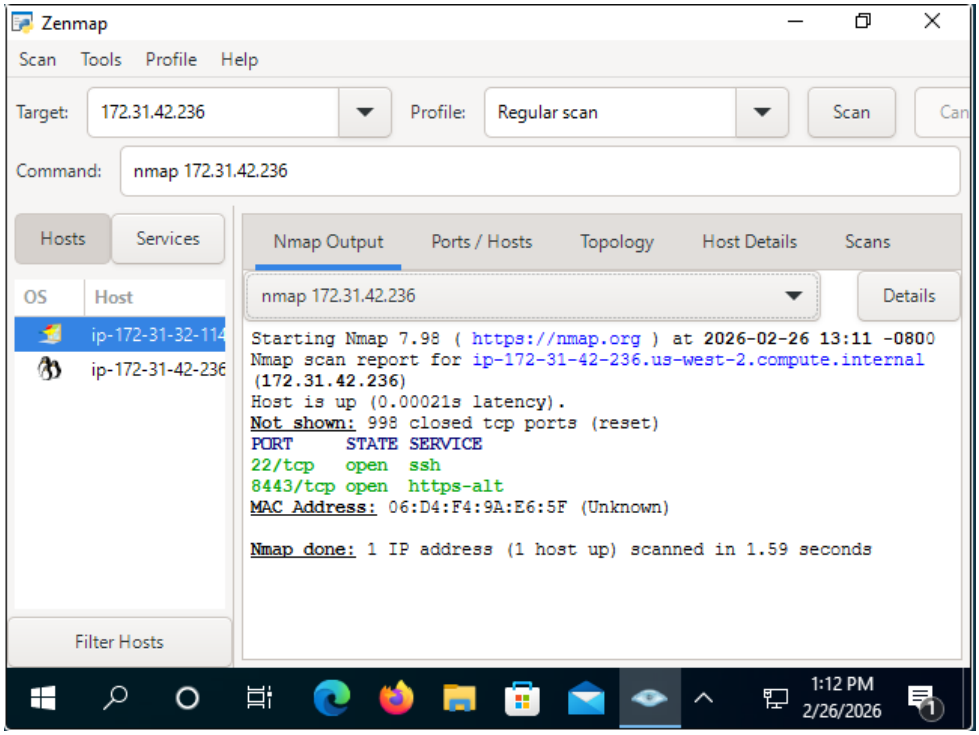
1.2 Debian Regular Scan

Command Used:
nmap 172.31.42.236

Open Ports Identified:

- 21/tcp – FTP
- 22/tcp – SSH
- 80/tcp – HTTP

Observation:
The host is reachable and critical services such as FTP, SSH, and HTTP are accessible.



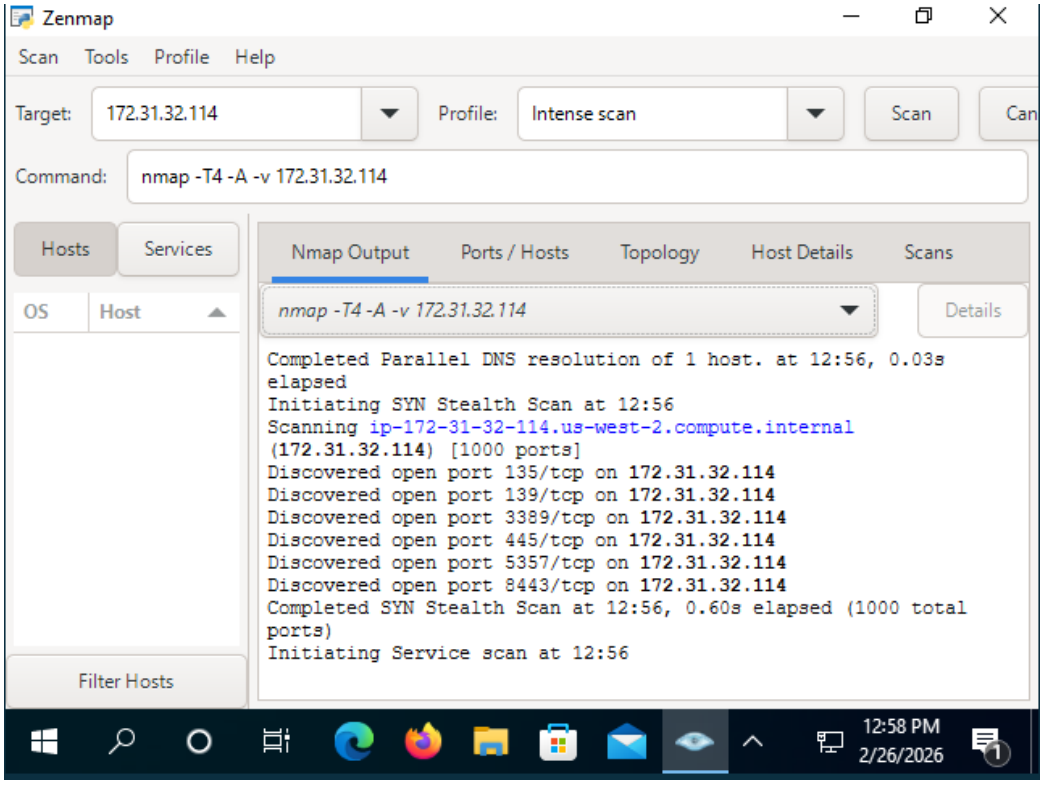
Step 2: Performing enumeration and service scanning

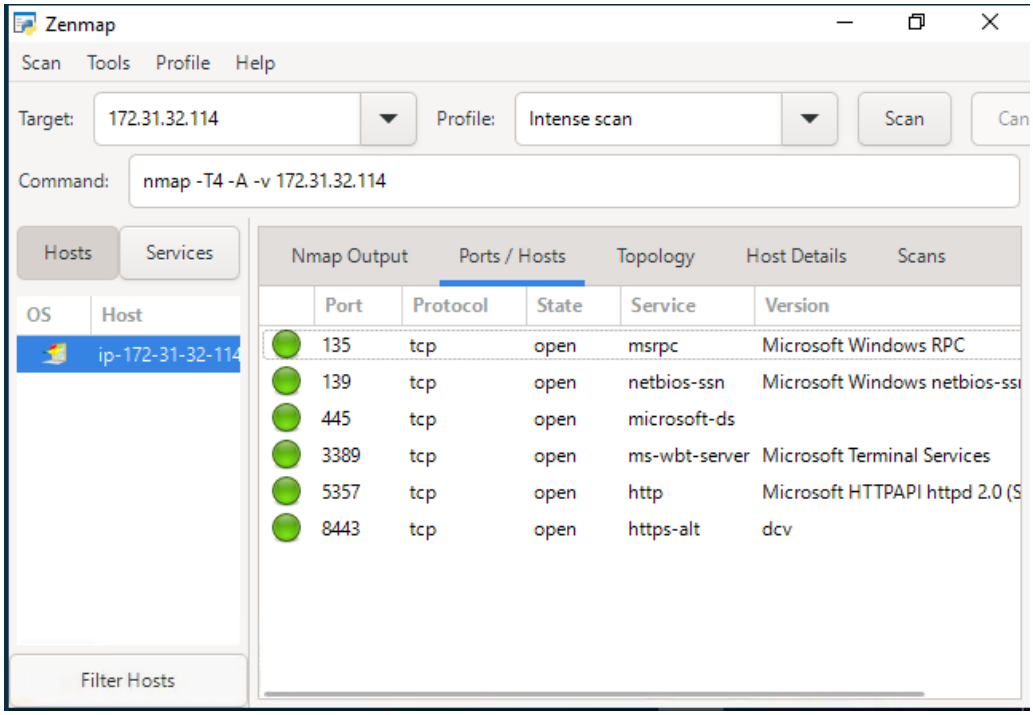
2.1 Windows Intense Scan

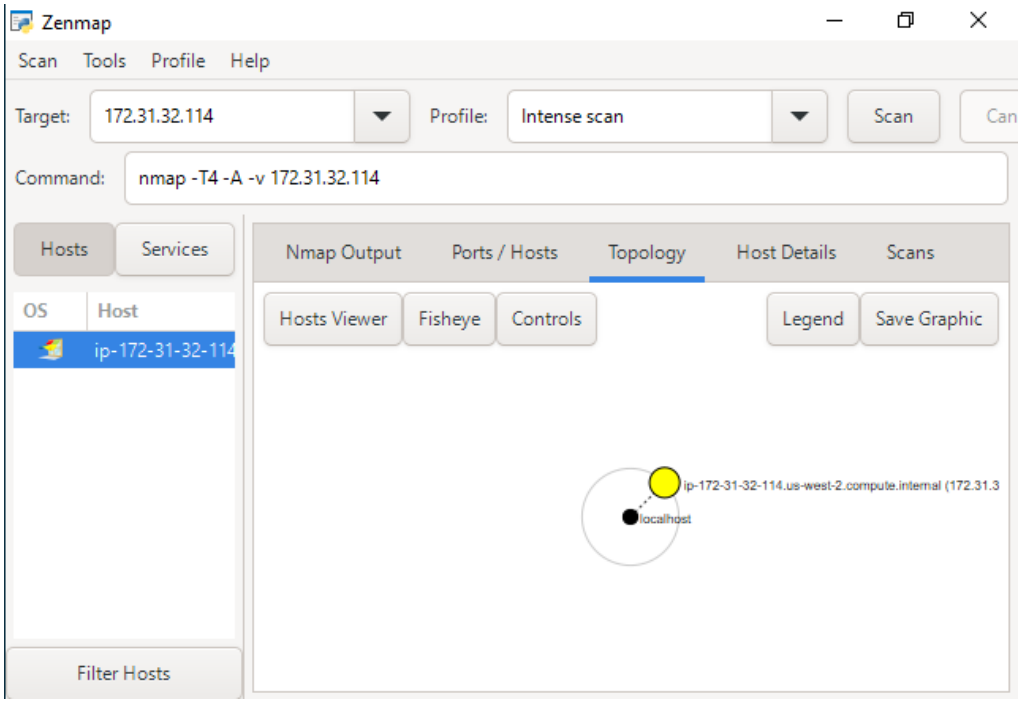
Command Used:
nmap -T4 -A -v 172.31.32.114

Findings:

- OS detected as Windows
- SMB service running
- RDP service active
- Service versions identified


Zenmap
Scan Tools Profile Help
Target: 172.31.32.114
Profile: Intense scan
Scan
Command: nmap -T4 -A -v 172.31.32.114
Hosts Services
OS Host
Filter Hosts
Nmap Output Ports / Hosts Topology Host Details Scans
nmap -T4 -A -v 172.31.32.114
Details
Completed Parallel DNS resolution of 1 host. at 12:56, 0.03s elapsed
Initiating SYN Stealth Scan at 12:56
Scanning ip-172-31-32-114.us-west-2.compute.internal (172.31.32.114) [1000 ports]
Discovered open port 135/tcp on 172.31.32.114
Discovered open port 139/tcp on 172.31.32.114
Discovered open port 3389/tcp on 172.31.32.114
Discovered open port 445/tcp on 172.31.32.114
Discovered open port 5357/tcp on 172.31.32.114
Discovered open port 8443/tcp on 172.31.32.114
Completed SYN Stealth Scan at 12:56, 0.60s elapsed (1000 total ports)
Initiating Service scan at 12:56
12:58 PM
2/26/2026


Zenmap
Scan Tools Profile Help
Target: 172.31.32.114
Profile: Intense scan
Scan
Command: nmap -T4 -A -v 172.31.32.114
Hosts Services
OS Host
ip-172-31-32-114
Filter Hosts
Nmap Output Ports / Hosts Topology Host Details Scans
Port Protocol State Service Version
135 tcp open msrpc Microsoft Windows RPC
139 tcp open netbios-ssn Microsoft Windows netbios-ssn
445 tcp open microsoft-ds
3389 tcp open ms-wbt-server Microsoft Terminal Services
5357 tcp open http Microsoft HTTPAPI httpd 2.0 (S
8443 tcp open https-alt dcv

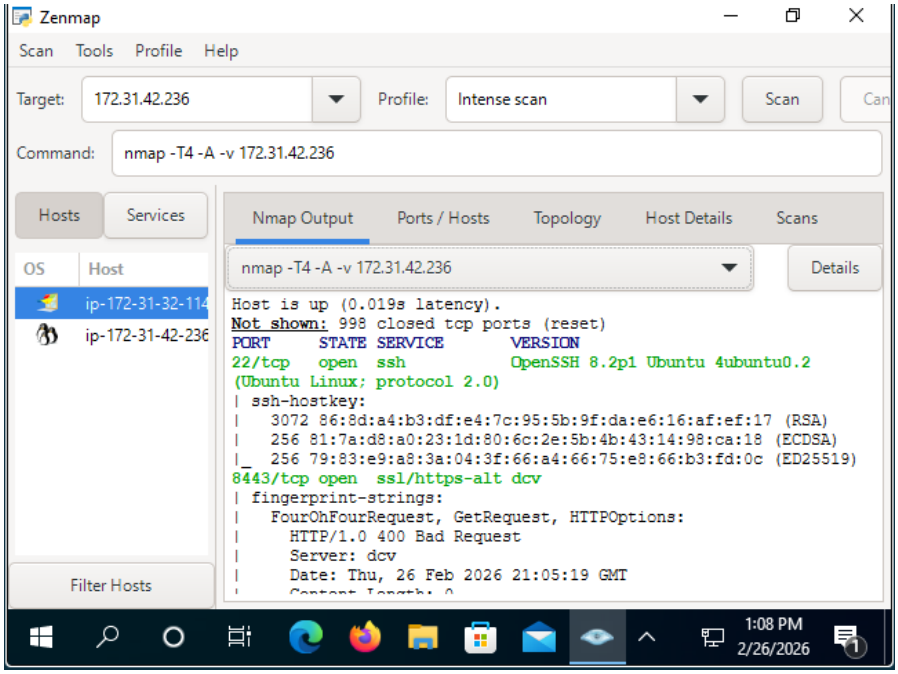


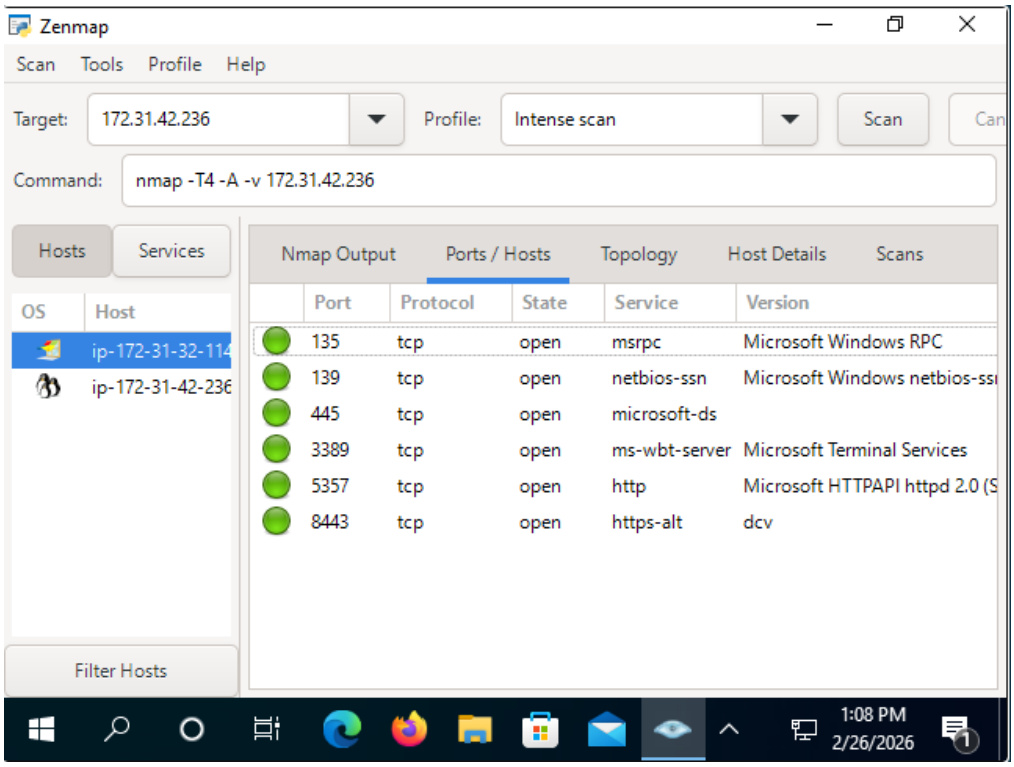
2.2 Debian Intense Scan

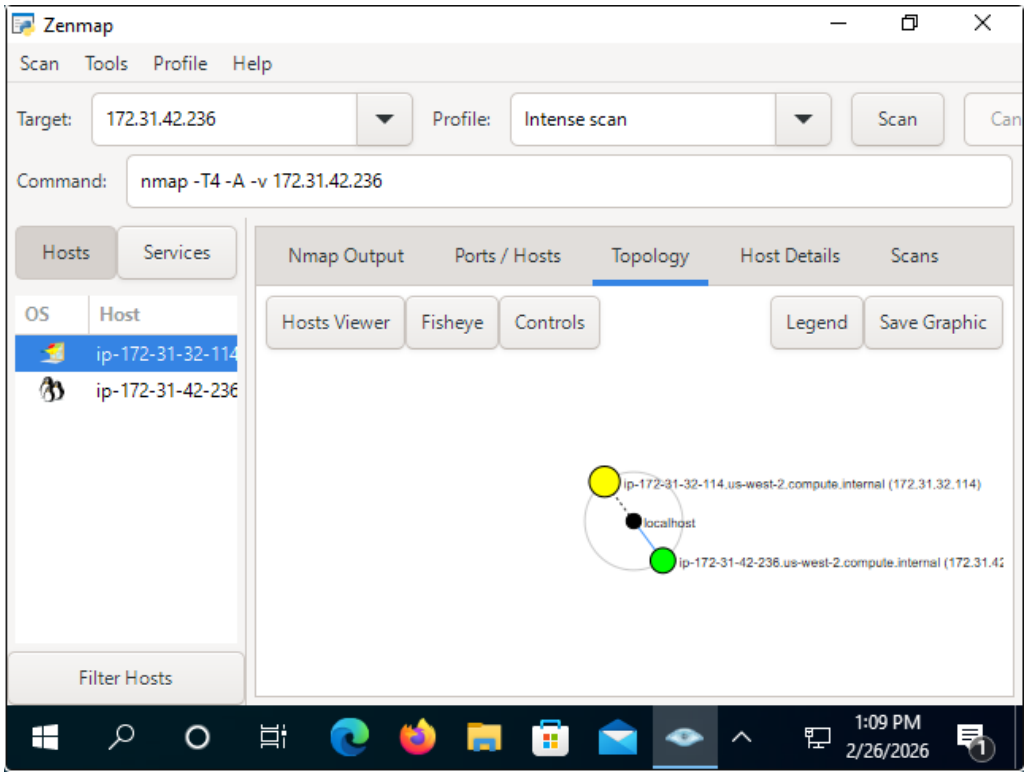
Command Used:
nmap -T4 -A -v 172.31.42.236

Findings:

- FTP service detected
- SSH service detected
- Apache HTTP server running
- OS identified as Debian Linux

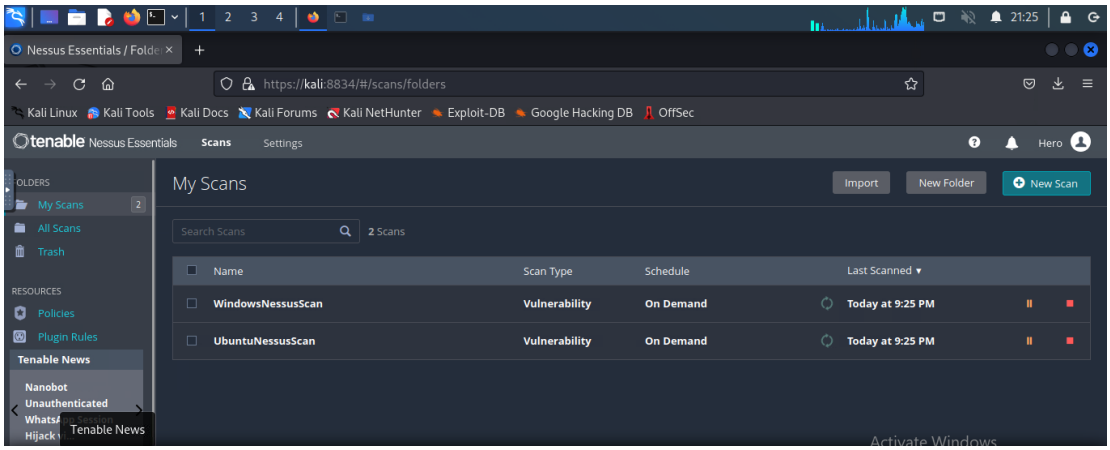



Zenmap
Scan Tools Profile Help
Target: 172.31.42.236
Profile: Intense scan
Scan
Command: nmap -T4 -A -v 172.31.42.236
Hosts Services
OS Host
ip-172-31-32-114
ip-172-31-42-236
Nmap Output Ports / Hosts Topology Host Details Scans
Port Protocol State Service Version
135 tcp open msrpc Microsoft Windows RPC
139 tcp open netbios-ssn Microsoft Windows netbios-ss
445 tcp open microsoft-ds
3389 tcp open ms-wbt-server Microsoft Terminal Services
5357 tcp open http Microsoft HTTPAPI httpd 2.0 (S
8443 tcp open https-alt dcv
Filter Hosts
1:08 PM
2/26/2026


Zenmap
Scan Tools Profile Help
Target: 172.31.42.236
Profile: Intense scan
Scan
Command: nmap -T4 -A -v 172.31.42.236
Hosts Services
OS Host
ip-172-31-32-114
ip-172-31-42-236
Nmap Output Ports / Hosts Topology Host Details Scans
Hosts Viewer Fisheye Controls Legend Save Graphic
ip-172-31-32-114.us-west-2.compute.internal (172.31.32.114)
localhost
ip-172-31-42-236.us-west-2.compute.internal (172.31.4
Filter Hosts
1:09 PM
2/26/2026

Step 3: Performing vulnerability assessment

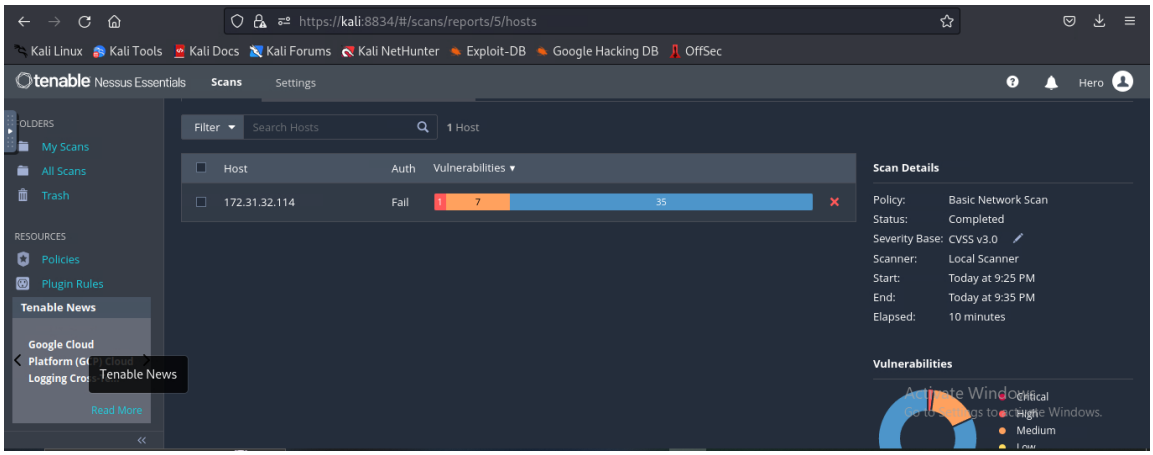
3.1 Nessus Scan – Windows

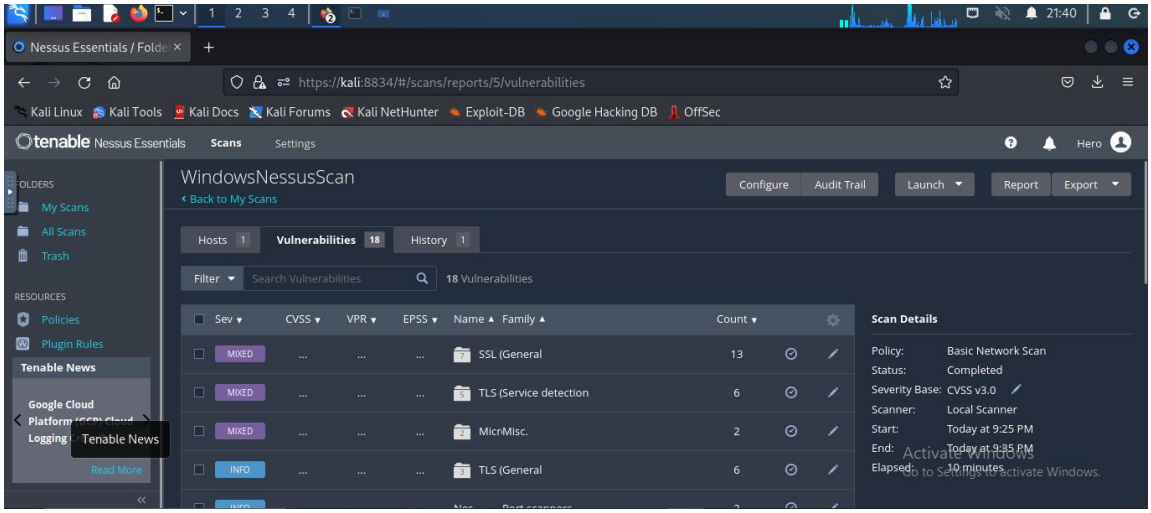


- Scan successfully completed
- High, Medium, and Low severity vulnerabilities identified

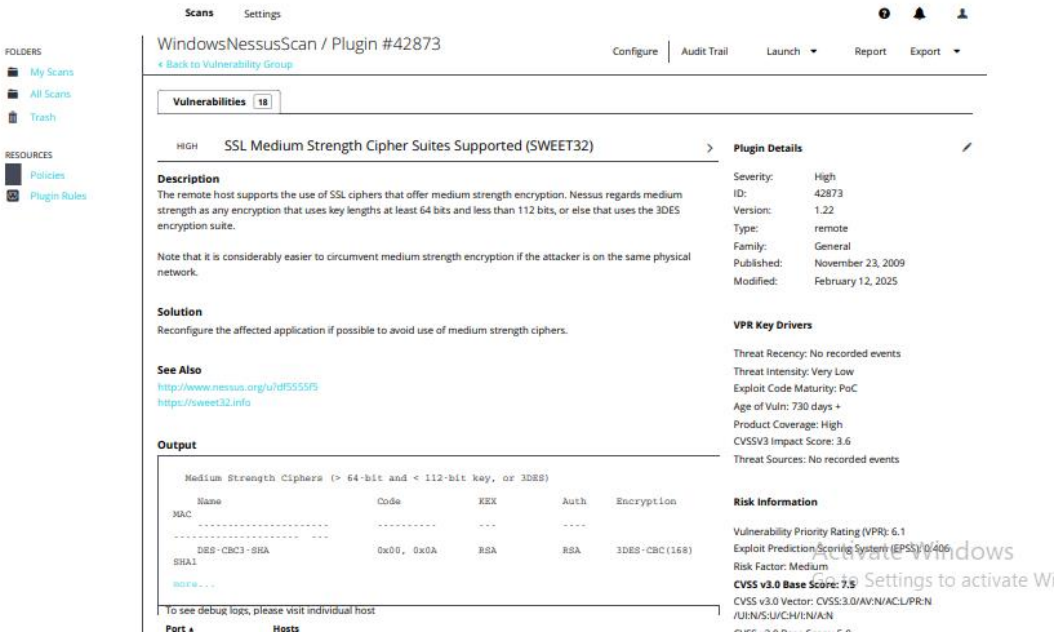
Common Observations:

- Missing security updates
- SMB configuration weaknesses
- RDP exposure





Nessus Essentials / Folders / View Scan https://kali:8834/#/scans/reports/5/hosts/2/vulnerabilit...

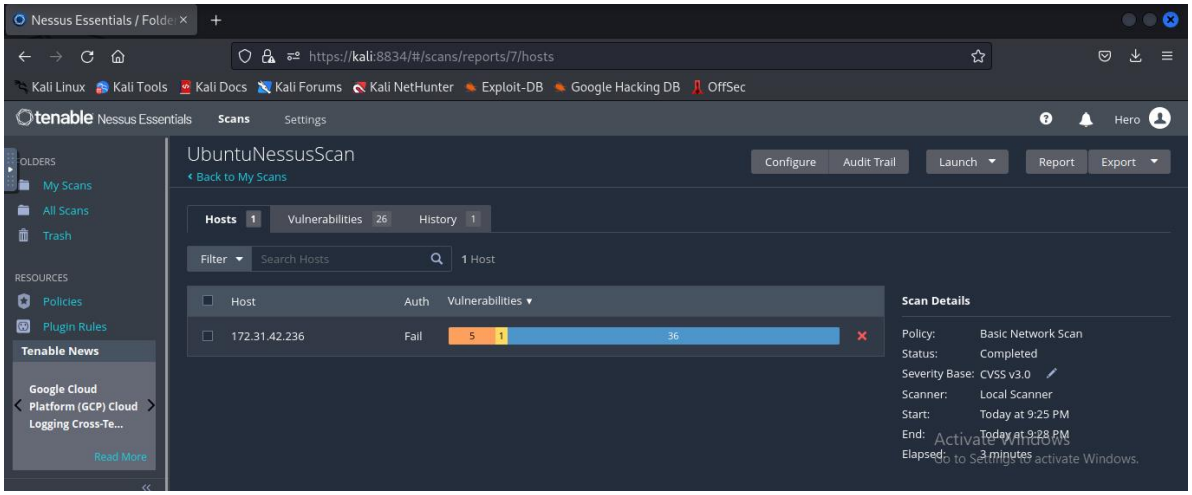


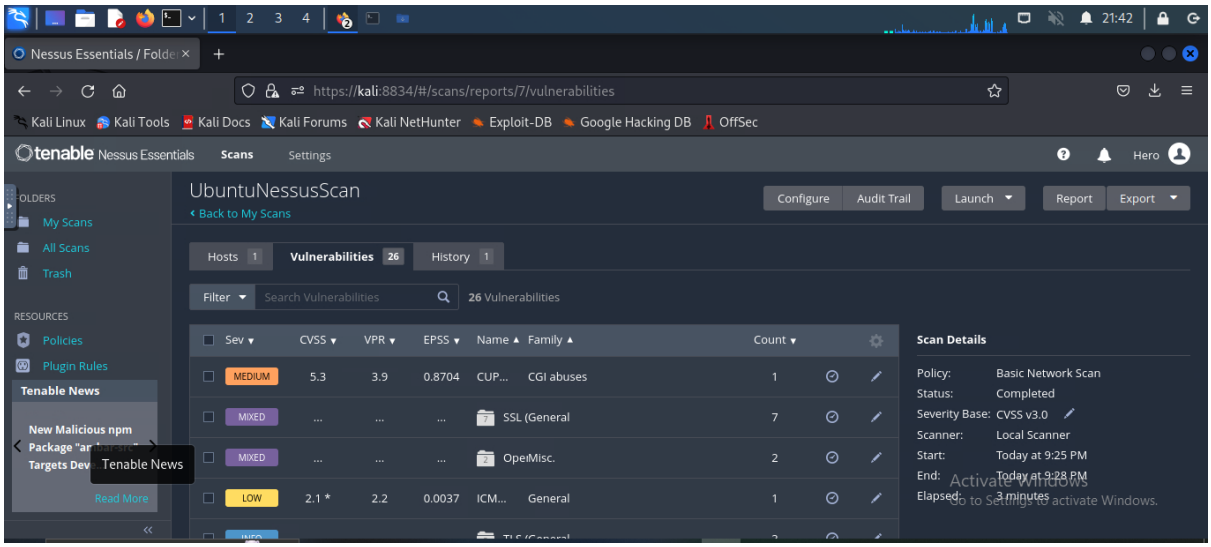
3.2 Nessus Scan – Debian

- Scan successfully completed
- Medium and Low severity vulnerabilities identified

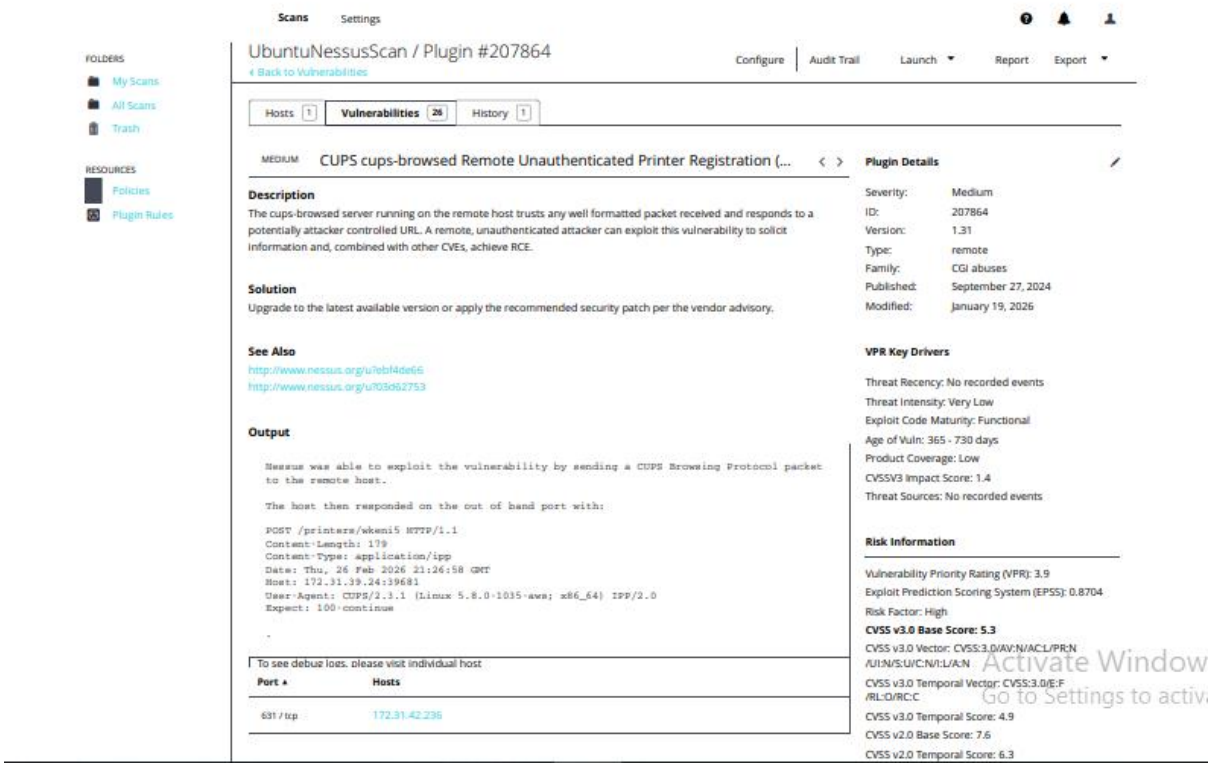
Common Observations:

- FTP exposure
- Outdated Apache packages
- Patch-related vulnerabilities


Nessus Essentials / Folde
https://kali:8834/#/scans/reports/7/hosts
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec
tenable Nessus Essentials
Scans
Settings
Hero
FOLDERS
My Scans
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Tenable News
Google Cloud Platform (GCP) Cloud Logging Cross-Te...
Read More
UbuntuNessusScan
Back to My Scans
Configure
Audit Trail
Launch
Report
Export
Hosts 1
Vulnerabilities 26
History 1
Filter
Search Hosts
1 Host
Host
Auth
Vulnerabilities
172.31.42.236
Fail
5
1
36
Scan Details
Policy: Basic Network Scan
Status: Completed
Severity Base: CVSS v3.0
Scanner: Local Scanner
Start: Today at 9:25 PM
End: Today at 9:28 PM
Elapsed: 3 minutes
Activate Windows
Go to Settings to activate Windows.


1 2 3 4
21:42
Nessus Essentials / Folde
https://kali:8834/#/scans/reports/7/vulnerabilities
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec
tenable Nessus Essentials
Scans
Settings
Hero
FOLDERS
My Scans
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Tenable News
New Malicious npm Package "ar
Targets Dev
Tenable News
Read More
UbuntuNessusScan
Back to My Scans
Configure
Audit Trail
Launch
Report
Export
Hosts 1
Vulnerabilities 26
History 1
Filter
Search Vulnerabilities
26 Vulnerabilities
Sev
CVSS
VPR
EPSS
Name
Family
Count
MEDIUM 5.3 3.9 0.8704 CUP... CGI abuses 1
MIXED ... ... SSL (General 7
MIXED ... ... OpenMisc. 2
LOW 2.1 * 2.2 0.0037 ICM... General 1
Scan Details
Policy: Basic Network Scan
Status: Completed
Severity Base: CVSS v3.0
Scanner: Local Scanner
Start: Today at 9:25 PM
End: Today at 9:28 PM
Elapsed: 3 minutes
Activate Windows
Go to Settings to activate Windows.

Nessus Essentials / Folders / View Scan https://kali:8834/#/scans/reports/7/vulnerabilities/207864



4. Vulnerability Summary Table

Operating System	IP Address	Vulnerability Count	Risk Classification	Reference
Windows	172.31.32.114	18	High	C:\Users\hp\Downloads\Nessus Essentials Folders View Scan.pdf
Debian	172.31.42.236	26	Medium–High	C:\Users\hp\Downloads\Nessus Essentials Folders View ScanU.pdf

5. Risk Analysis

Windows Risks:

- SMB exploitation possibility
- RDP brute-force risk
- Privilege escalation exposure

Debian Risks:

- Unauthorized FTP access
- SSH brute-force possibility
- Web server exposure

6. Conclusion

The VAPT assessment identified several vulnerabilities across both systems.
The Windows system presents higher risk due to exposed SMB and RDP services along with patch gaps.
The Debian system presents moderate risk due to exposed network services and outdated components.

Remediation of identified vulnerabilities is recommended to reduce the overall attack surface.

7. Executive Summary:-

A Vulnerability Assessment and Penetration Testing (VAPT) exercise was conducted on two internal systems to evaluate their security posture and identify potential weaknesses.

The assessment revealed multiple vulnerabilities, including high-risk exposures that could lead to unauthorized access, data compromise, or service disruption if exploited.

The purpose of this assessment was to proactively identify and address security gaps before they can be leveraged by malicious actors.

8. Executive Report (For Senior Leadership)

Assessment Overview

The following systems were assessed:

- Windows – 172.31.32.114
- Debian – 172.31.42.236

The evaluation focused on identifying exposed services, configuration weaknesses, and missing security patches.

Key Risks Identified

Windows System

- SMB and RDP services exposed
- Missing patches
- Configuration weaknesses

Business Impact:

- Risk of ransomware
- Unauthorized access
- Data compromise

Debian System

- FTP and SSH exposed
- Web server vulnerabilities
- Outdated packages

Business Impact:

- Unauthorized remote login
- Credential compromise
- Service disruption

Executive Risk Dashboard

System	Risk Level	Primary Exposure	Business Impact
Windows	High	SMB, RDP	Data breach risk
Debian	Medium–High	FTP, SSH	Unauthorized access

9. Technical Report (For Technical Team)

Scope

Targets Assessed:

System	IP Address
Windows	172.31.32.114
Debian	172.31.42.236

Tools Used:

- Zenmap
- Nessus

Technical Findings Summary

Windows

- Multiple open ports including SMB and RDP
- Patch-related vulnerabilities detected
- High severity findings present

Debian

- FTP, SSH, and HTTP exposed
- Outdated packages identified
- Medium severity findings present